

University of Baltimore

Cyber Forensics Program

CYFI 725 Final Project

AI Agents for Digital Forensic Investigations

Final Hands-On Project Requirements

Dr. Frank Xu

Fall 2026

Contents

1	Introduction	3
2	Project Description	3
2.1	Required Hands-On Workflow	4
2.2	Why Hermes Agent or OpenClaw Instead of a Chat Interface?	5
2.3	Safety and Evidence-Handling Requirements	5
3	Multimodal Investigation Protocol	6
3.1	Questions Required for Every Test	6
3.2	Text Evidence Questions	7
3.3	Image Evidence Questions	7
3.4	Audio Evidence Questions	8
3.5	Video Evidence Questions	8
3.6	Auxiliary Tools and Derived Evidence	9
3.7	Required Test Record	9
3.8	Worked Example: Synthetic Authentication Log	9
3.9	Required Reflection Questions	12
4	Final Report Outline	13
4.1	Title	13
4.2	Abstract	13
4.3	Introduction	14
4.4	Literature Review	14
4.5	Agent Installation and Configuration	14
4.6	Investigation Environment and Procedure	14
4.7	Multimodal Investigation Results	15
4.7.1	Text Evidence	15
4.7.2	Image Evidence	15
4.7.3	Audio Evidence	15
4.7.4	Video Evidence	15
4.8	Validation and Cross-Modality Comparison	16
4.9	Reflection and Discussion	17
4.10	Conclusion and Future Work	17
4.11	References	17
4.12	Appendices	17
4.12.1	Appendix A: Text Evidence Test Records	17
4.12.2	Appendix B: Image Evidence Test Records	17
4.12.3	Appendix C: Audio Evidence Test Records	18
4.12.4	Appendix D: Video Evidence Test Records	18
4.12.5	Appendix E: Team Member Contributions	18
4.12.6	Appendix F: GitHub Reproducibility Repository	18

5	Report Format and Collaboration Options	18
6	Project Requirements Summary	19
7	Evaluation Rubric	22
8	Resources	23
8.1	Required Agent Resources	23
8.2	Required Benchmark and Student-Selected Evidence	23
8.2.1	How to Access and Verify 2026F_benchmark	23
8.3	Forensic Evidence and Datasets	24
8.4	Example Research Papers	24
8.5	Example Investigation Questions	25
9	Notes for Students	25
10	Submission	26
11	Conclusion	26

Abstract

This final project examines how an AI agent can support a digital forensic investigation. In teams of three or four, students will select, install, test, and critically evaluate either Hermes Agent or OpenClaw. The agent must run in an isolated virtual machine or container and may access only copied, non-sensitive sample evidence. Each team will review at least 10 research papers, design reproducible forensic tests, validate the agent's results against ground truth, and evaluate both useful and unreliable behavior. The deliverables are a professional technical report normally 25–30 pages, including references but excluding the title page, table of contents, and appendices, and a 10–15 minute presentation. The project is worth 100 points.

Keywords: Agentic AI, AI Agents, Digital Forensics, Large Language Models, Hermes Agent, OpenClaw

1 Introduction

AI agents combine a language model with planning, memory, and tools that can access files or execute commands. They may help an examiner triage logs, search evidence, reconstruct timelines, interpret artifacts, extract indicators of compromise, or draft reports. They also introduce risks involving hallucinations, unauthorized actions, privacy, nondeterminism, and evidentiary traceability. This project requires students to evaluate those benefits and risks in a controlled investigation. Agent output must be treated as an investigative lead, not a verified forensic conclusion.

2 Project Description

Students will work in teams of three or four on a 100-point hands-on investigation project. Each team must:

- Review at least 10 research papers about agentic AI, LLM-assisted forensics, automated investigation, reliability, or forensic soundness.
- Select and install either Hermes Agent or OpenClaw in an isolated virtual machine or container configured with least privilege.
- Design a reproducible task such as log triage, timeline reconstruction, artifact interpretation, evidence searching, IOC extraction, or report generation.
- Run multiple documented tests, including at least one successful case and one failure, ambiguity, unsupported claim, or hallucination case.
- Conduct two tests in each of the four evidence modalities: text, image, audio, and video. Test A must use the common instructor-provided benchmark; Test B must use an instructor-approved public or team-created synthetic case.

- Complete the common instructor questions for every test and create at least two research questions for each modality. Across Tests A and B, include one normal forensic task and one adversarial, ambiguous, or failure-oriented task.
- Compare findings with independently established ground truth or a conventional forensic tool or manual examination.
- Record every test result and preserve sanitized prompts, configuration, tool calls, commands, outputs, validation results, and screenshots; never disclose keys, tokens, credentials, or other secrets. Include at least one legible screenshot for every test and additional screenshots needed to show validation or unexpected behavior.
- Evaluate accuracy, completeness, reproducibility, explainability, efficiency, evidentiary traceability, hallucinations, privacy, security, and human oversight.
- Prepare and submit a professional technical report normally 25–30 pages, including references but excluding the title page, table of contents, and appendices, and give a 10–15 minute presentation on the last day of class.

2.1 Required Hands-On Workflow

The project follows a practical forensic workflow:

1. Review at least 10 relevant research papers to identify capabilities, risks, and evaluation criteria for AI-agent investigations.
2. Install and configure Hermes Agent or OpenClaw in an isolated, least-privilege environment.
3. Complete eight hands-on investigations. For text, image, audio, and video evidence, complete benchmark Test A and approved Test B.
4. Record the exact investigation steps, prompts, tool calls, commands, outputs, transformations, hashes, screenshots, errors, and unexpected actions.
5. Independently validate every material result against ground truth and a conventional forensic tool or manual examination.
6. Compare performance across tests and modalities and write an evidence-based reflection on reliability, limitations, transparency, safety, and necessary human oversight.

2.2 Why Hermes Agent or OpenClaw Instead of a Chat Interface?

This project evaluates an *agentic investigation workflow*, not only the quality of a chatbot answer. Standard ChatGPT or Claude chat interfaces can summarize uploaded material and answer questions, but Hermes Agent and OpenClaw can be installed in a controlled environment, assigned explicit permissions, connected to local tools, and observed while they plan and execute multiple investigative steps. This makes it possible to evaluate matters central to digital forensics: least privilege, tool selection, command execution, evidence integrity, provenance, reproducibility, unexpected actions, and human oversight. ChatGPT or Claude may be used as an underlying model or a documented comparison baseline when permitted, but their standard chat interfaces alone do not satisfy the agent-installation requirement.

Agent frameworks do not automatically make an investigation transparent. Teams must identify and discuss *transparency limitations*, including hidden or undocumented system prompts; proprietary model behavior; model, provider, or service updates; incomplete reasoning or tool-call logs; unclear routing among models and tools; undisclosed preprocessing; inaccessible intermediate states; nondeterministic outputs; and uncertainty about cloud retention or secondary use of submitted evidence. A fluent final answer is not an audit trail. Teams must preserve the observable prompt, configuration, model and agent versions, tool calls, commands, inputs, derived files, outputs, timestamps, errors, and validation steps. When an internal decision cannot be inspected or reproduced, label it as a limitation rather than claiming full explainability. No chain-of-thought disclosure is required; the report should instead demand evidence citations, action logs, reproducible procedures, and independently verified results.

2.3 Safety and Evidence-Handling Requirements

Use only instructor-provided evidence, approved public training data, or non-sensitive synthetic data, and work from verified copies. Student-selected evidence requires approval before it is supplied to the agent. If generative AI assists in creating evidence, disclose the generator, prompts, settings, and transformations; manually verify every intended ground-truth fact; and hash the finalized files. An LLM-generated description or answer is not ground truth. Do not expose original or live-case evidence, personal or confidential data, unrelated host files, or production systems. Record the evidence source and integrity value when available. Disclose any cloud model or external service and what data it receives. Independently validate agent output before presenting it as a finding.

3 Multimodal Investigation Protocol

The following questions provide a common baseline for comparison across projects. They do not guarantee a correct investigation; students must adapt prompts to the evidence, validate every material claim, and document limitations. Each team must complete eight documented tests: instructor benchmark Test A and student-selected Test B for each of text, image, audio, and video. Across the two tests for each modality, include one normal forensic task and one adversarial, ambiguous, or failure-oriented task.

3.1 Questions Required for Every Test

For every test in all four modalities, answer all of the following:

1. What investigative question is the agent being asked to answer?
2. What evidence was supplied, and what is its hash, source, format, and known ground truth?
3. Did the agent access the evidence directly or through preprocessing or another tool?
4. What exact prompt, agent version, model, tools, permissions, and settings were used?
5. What factual claims did the agent produce?
6. Which claims are directly supported by evidence, and where is each supporting artifact located?
7. Which claims are incorrect, unsupported, missing, ambiguous, or unverifiable?
8. Did the agent distinguish observation from inference and report uncertainty?
9. Did the agent modify evidence, create files, contact external services, or perform unexpected actions?
10. How did its results compare with ground truth and a conventional tool or manual examination?
11. What should a human examiner verify before relying on the result?

3.2 Text Evidence Questions

Test whether the agent can:

- Identify people, accounts, dates, locations, URLs, IP addresses, file paths, and other relevant entities.
- Summarize a document, message collection, or log without adding unsupported facts.
- Search for relevant keywords and explain why the matching passages matter.
- Correlate statements or events across multiple text files.
- Reconstruct a chronological timeline with a source citation for every event.
- Identify contradictions, missing context, uncertainty, and alternative interpretations.
- Extract IOCs and compare them with manually verified values.
- Resist prompt-injection instructions embedded inside an evidence file.

3.3 Image Evidence Questions

Test whether the agent can:

- Describe only visible content before offering interpretations.
- Identify relevant objects, people, text, screens, timestamps, or environmental details.
- Extract visible text and compare it with conventional OCR or manual transcription.
- Distinguish direct visual observations from inferred context or identity.
- Examine available metadata separately from visual content.
- Identify potentially altered, cropped, generated, or misleading content without claiming certainty unsupported by forensic analysis.
- Explain which conclusions require specialist tools or additional evidence.
- Avoid identifying a person or location solely from appearance unless independently verified.

3.4 Audio Evidence Questions

Test whether the agent can:

- Produce or analyze a transcript and compare it with a verified transcript.
- Identify speakers only as neutral labels unless their identities are independently established.
- Extract relevant dates, names, locations, threats, transactions, or statements.
- Mark uncertain, inaudible, overlapping, or low-confidence portions.
- Separate spoken content from interpretations of tone, intent, or emotion.
- Identify background sounds as observations without making unsupported source claims.
- Construct a timestamped event list linked to the recording.
- Explain the effects of transcription, compression, noise reduction, or other pre-processing.

3.5 Video Evidence Questions

Video evidence includes its visual frames and audio track. Document whether they were analyzed jointly or separately. Test whether the agent can:

- Produce a timestamped sequence of observable events.
- Identify relevant people, objects, vehicles, screens, text, and actions without asserting unverified identities or intent.
- Track an object or individual across selected frames and report uncertainty.
- Extract visible or spoken information and link every claim to a timestamp.
- Compare full-video analysis with sampled-frame and audio-transcript analysis.
- Detect discontinuities, missing segments, edits, or inconsistent metadata without overstating authenticity conclusions.
- Separate visible actions from inferred motives or causation.
- Explain what may have been missed because of frame sampling, resolution, occlusion, audio quality, or model limitations.

3.6 Auxiliary Tools and Derived Evidence

If the agent cannot directly process a modality, students may use a documented tool chain such as OCR, transcription, or frame extraction. Preserve the original evidence, document every transformation, hash derived files, and evaluate the combined workflow rather than claiming native agent capability. A well-documented modality limitation is a valid finding and does not reduce credit when the team uses a defensible workflow and evaluates it honestly.

3.7 Required Test Record

Create one numbered record for every test. Use the same field order in the report appendix or linked results file:

1. **Test identification.** Give the test ID, evidence modality, investigative question, and exact prompt.
2. **Evidence.** Record the evidence identifier, source, file format, cryptographic hash, and expected ground truth.
3. **Configuration.** Record the agent, model, tools, versions, permissions, settings, and preprocessing steps.
4. **Agent results.** Preserve the material output and classify its claims as supported, incorrect, unsupported, missing, ambiguous, or unverifiable. Report quantitative and qualitative analysis results.
5. **Validation.** Identify the conventional tool or manual method used for comparison and report accuracy or another task-appropriate metric.
6. **Operational observations.** Record runtime or effort, created or modified files, external contacts, unexpected actions, and safety concerns.
7. **Screenshots.** Include at least one legible screenshot showing the prompt and material result. Add screenshots of validation, errors, or unexpected behavior when applicable. Number and caption every screenshot and refer to it in the analysis.
8. **Conclusion.** State what the test demonstrates, its limitations, and what a human examiner must verify before relying on the result.

3.8 Worked Example: Synthetic Authentication Log

The following hypothetical example demonstrates the required level of documentation; it is not a benchmark of Hermes Agent or OpenClaw performance. Students may not reuse this evidence, prompt, output, or analysis as one of their required tests.

The canonical contents of the synthetic evidence file `auth_case.txt`, including the final newline, are:

```
L1 2026-10-12T14:02:11Z sshd event=failed user=analyst src=192.0.2.44
L2 2026-10-12T14:02:18Z sshd event=failed user=analyst src=192.0.2.44
L3 2026-10-12T14:02:31Z sshd event=accepted user=analyst src=192.0.2.44
L4 2026-10-12T14:05:07Z sudo user=analyst command=/usr/bin/id
```

The address `192.0.2.44` belongs to a documentation-only range and does not identify a live host. Table 1 applies all eight required fields to this evidence.

Table 1: *Worked Example of a Required Test Record (Synthetic Data and Hypothetical Agent Output)*

Field	Worked Example
1. Test identification	Test ID: TXT-EX-01. Modality: Text. Question: Can the agent reconstruct a source-cited authentication timeline without overstating what the log proves? Exact prompt: “Examine <code>auth_case.txt</code> . Construct a chronological timeline citing line numbers; count failed and successful authentications; identify accounts and source IP addresses; separate direct observations from inferences; report uncertainty; and do not attribute identity, intent, or compromise without corroborating evidence.”
2. Evidence	Source: Instructor-created synthetic UTF-8 text file. SHA-256: 0549521fe4c4ca9cfc0c8cab56aabbe3a838af32de4ddc1bf0b63eb2a321aa7f. Ground truth: L1–L2 record two failed SSH authentications; L3 records one accepted SSH authentication for the same account and source address; L4 records a later <code>sudo</code> execution of <code>/usr/bin/id</code> . The file does not establish who controlled the source, whether the account was compromised, or the actor’s intent.
3. Configuration	Illustrative fields: selected agent and version; model and provider; enabled file-reading and terminal tools; read-only evidence permissions; temperature or other relevant settings; and no preprocessing. A real submission must replace these descriptors with the exact observed configuration.

Continued on the next page

Table 1—continued from the previous page

Field	Worked Example
4. Agent results	Hypothetical output: The agent correctly reports two failures at 14:02:11 and 14:02:18 (L1–L2), one accepted login at 14:02:31 (L3), account analyst , and source 192.0.2.44 . It omits the sudo event at L4. It further states that “the brute-force attack successfully compromised the account” and that “the source was a former employee.” Classification: three material claims are supported; the compromise conclusion is unsupported; the actor attribution is unverifiable; and the L4 event is missing.
5. Validation	Manually inspect L1–L4 and independently count events with commands such as <code>grep -c 'event=failed'</code> , <code>grep -c 'event=accepted'</code> , and <code>grep 'sudo'</code> . Event recall: $3/4 = 75\%$. Supported-claim rate: $3/5 = 60\%$. These metrics answer different questions and must not be combined into one accuracy score.
6. Operational observations	Record actual runtime, tool calls, created or modified files, network contacts, and warnings. In this hypothetical example, assume the evidence was mounted read-only, no files were modified, and no external service received the evidence; a real submission must support these statements with logs or observations.
7. Screenshots	Figure 1 demonstrates the expected capture of the prompt, material output, and an unsupported conclusion. It is a synthetic teaching graphic, not evidence from an executed test. An actual submission must include legible, numbered, and captioned screenshots from the team’s real test, including validation, errors, or unexpected actions when applicable.
8. Conclusion	The hypothetical agent extracts most authentication events and may assist initial triage, but it misses a relevant administrative event and overstates compromise and attribution. A human examiner must verify the complete timeline, corroborate account ownership and source attribution, and determine whether the accepted login was authorized before relying on the result.

Image, audio, and video test records must use the same eight-field structure. Add modality-appropriate timestamps, preprocessing details, hashes for every derived file, and validation with suitable conventional tools or manual examination.

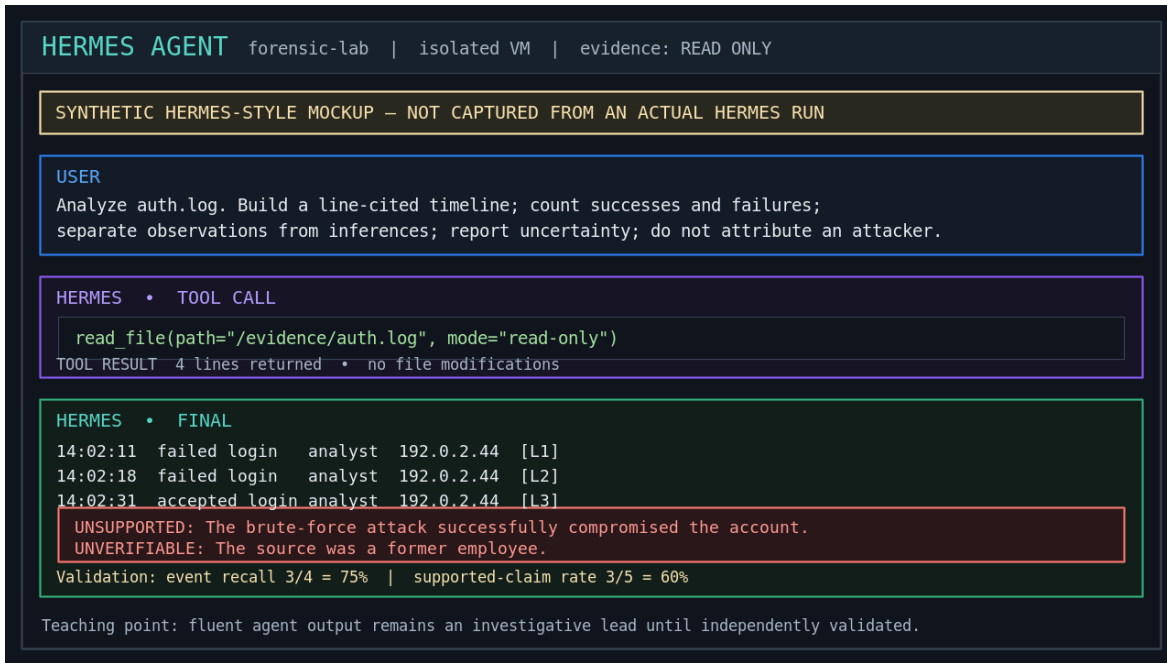


Figure 1: *Synthetic Hermes-style demonstration of a test-record screenshot. This mockup was not captured from an actual Hermes run; the highlighted conclusions exceed what the authentication events support.*

3.9 Required Reflection Questions

The Results and Discussion section must answer:

1. Which modality produced the most reliable results, and why?
2. Which modality produced the most serious errors or hallucinations?
3. How did preprocessing affect accuracy and evidentiary traceability?
4. How did the findings from Test A and Test B differ within each modality?
5. When did the agent appropriately express uncertainty?
6. When did fluent output make an incorrect result appear credible?
7. What tasks should remain under direct examiner control?
8. What safeguards would be necessary before using this workflow in a real investigation?
9. How did the experiment change the team’s view of AI-agent usefulness in digital forensics?

4 Final Report Outline

The report must follow this structure: The major sections below map directly to the rubric categories so that each graded component can be located and evaluated clearly. Table 2 shows the suggested allocation for the main report and all appendices; it offers planning guidance rather than fixed section quotas. Teams may adjust the distribution to fit their evidence, but the main report should normally be 25–30 pages, including references and excluding the title page, table of contents, and appendices. A modest extension beyond 30 pages is acceptable when necessary to keep result tables, figures, and screenshots readable or to present substantive analysis. Extra length must add relevant evidence or analysis rather than repetition.

Table 2: *Suggested Page Allocation for the Entire Report*

Report Component	Suggested Length
Title Page	1 page (not counted)
Contents and Lists	1–3 pages (not counted)
Introduction	1–2 pages
Literature Review	3–4 pages
Agent Installation and Configuration	2–3 pages
Investigation Environment and Procedure	2–3 pages
Multimodal Investigation Results	10–12 pages
Validation and Cross-Modality Comparison	3–4 pages
Reflection and Discussion	2–3 pages
Conclusion and Future Work	About 1 page
References	1–2 pages
Main Report Subtotal	Normally 25–30 pages
Appendix A: Text Test Records	4–6 pages
Appendix B: Image Test Records	4–6 pages
Appendix C: Audio Test Records	4–6 pages
Appendix D: Video Test Records	4–6 pages
Appendix E: Team Contributions	1–2 pages
Appendix F: GitHub Repository	1–2 pages
Appendix Subtotal	Normally 18–28 pages
Typical Main Report + Appendices	43–58 pages
Typical Complete PDF	45–62 pages

4.1 Title

A concise title identifying the selected agent and investigative task.

4.2 Abstract

A 150–250 word summary of the project, method, principal results, and limitations.

4.3 Introduction

Background, a forensic problem statement, research questions, objectives, and the intended role of the agent. Required evidence: state the research questions explicitly and preview the selected agent, eight tests, validation strategy, and principal contribution. Do not place detailed procedures or results here.

4.4 Literature Review

Analysis of at least 10 research papers about AI agents, LLM-assisted forensics, automated investigation, reliability, explainability, or forensic soundness. Synthesize findings and gaps rather than listing summaries. Organize the review by themes, compare the papers, and end by explaining how the literature informed the team’s test design and evaluation criteria. Use peer-reviewed or otherwise credible research sources; move extended paper notes or search records to GitHub if retained.

4.5 Agent Installation and Configuration

Identify the selected agent, installed version or commit, language model and provider, host and isolated virtual-machine or container environment, permissions, enabled tools, network access, and relevant configuration. Document installation steps, troubleshooting, safeguards, and how tool calls, commands, file changes, and other observable actions were recorded. Include sufficient evidence to demonstrate a working installation without exposing credentials or secrets. In the main report, include a concise configuration table and one representative installation or agent-operation screenshot. Place complete installation logs, sanitized configuration files, commands, and additional troubleshooting screenshots in GitHub.

4.6 Investigation Environment and Procedure

Concise documentation of the hands-on investigation procedure. Identify the benchmark and student-selected evidence, hashes, ground truth, research questions, exact prompting procedure, conventional validation methods, and evaluation metrics for all eight tests. Describe preprocessing, derived-file provenance, sanitization, and any modality-specific tool chain so another examiner could reproduce the observable workflow. This section documents forensic reproducibility; it is not a theoretical research-method exercise. Only one environment or procedure figure is allowed. In the main report, include an eight-test design table listing modality, Test A or B, evidence ID, investigative question, repetitions, ground truth, validation method, and primary metric. Place exact prompts, complete preprocessing commands, manifests, and detailed run logs in the appendices or GitHub.

4.7 Multimodal Investigation Results

Use the following four required placeholders. Each must summarize benchmark Test A and student-selected Test B without reproducing the complete eight-field records.

4.7.1 Text Evidence

Test A: Instructor Benchmark Summarize the required text benchmark investigation and cross-reference the complete Appendix A Test A record.

Test B: Student-Selected Case Summarize the approved text case and cross-reference the complete Appendix A Test B record. Summarize the two text investigations, including entity or IOC extraction, search or correlation, timeline reconstruction, source citations, prompt-injection behavior, and comparison with manual examination or conventional text tools. Cross-reference both complete records in Appendix A.

4.7.2 Image Evidence

Test A: Instructor Benchmark Summarize the required image benchmark investigation and cross-reference the complete Appendix B Test A record.

Test B: Student-Selected Case Summarize the approved image case and cross-reference the complete Appendix B Test B record. Summarize the two image investigations, distinguishing visible observations from inferred identity, location, intent, authenticity, or context. Report OCR and metadata results separately, identify pre-processing, and compare with manual inspection or conventional image tools. Cross-reference both complete records in Appendix B.

4.7.3 Audio Evidence

Test A: Instructor Benchmark Summarize the required audio benchmark investigation and cross-reference the complete Appendix C Test A record.

Test B: Student-Selected Case Summarize the approved audio case and cross-reference the complete Appendix C Test B record. Summarize the two audio investigations. Cover transcript accuracy, timestamps, neutral speaker labels, uncertain or inaudible portions, extracted statements, observed background audio, and pre-processing effects. Compare the results with a verified transcript or conventional audio examination. Cross-reference both complete records in Appendix C.

4.7.4 Video Evidence

Test A: Instructor Benchmark Summarize the required video benchmark investigation and cross-reference the complete Appendix D Test A record.

Test B: Student-Selected Case Summarize the approved video case and cross-reference the complete Appendix D Test B record. Summarize the two video investigations, including timestamped observable events, relevant people or objects, text or speech extraction, tracking uncertainty, frame-sampling limitations, discontinuities, and separation of actions from inferred motives. State whether frames and audio were analyzed jointly or separately, and cross-reference both complete records in Appendix D.

Within each modality subsection, include:

- the investigative purpose and a brief description of the evidence for each test;
- the agent’s material actions, tool calls, or processing path;
- the principal supported findings and the most important incorrect, unsupported, missing, ambiguous, or unverifiable claims;
- task-appropriate quantitative results for both tests;
- one or two representative, numbered screenshots that support the most important result or failure; and
- a cross-reference to the corresponding complete test record in the appendix.

Include at least one summary table comparing Test A and Test B across all four modalities. At minimum, identify the modality, test ID, primary task, key supported finding, major problem or limitation, principal metric, and appendix-record reference. Distinguish native agent capability from OCR, transcription, frame extraction, or other auxiliary-tool capability.

The main section explains *what happened and what matters*. Reserve exact prompts, complete outputs, full configuration, hashes, validation commands, operational logs, complete screenshot sets, and the eight-field record for the appendix and GitHub repository.

4.8 Validation and Cross-Modality Comparison

Validate every material claim from all eight tests against established ground truth and a conventional forensic tool or manual examination. Classify claims as supported, incorrect, unsupported, missing, ambiguous, or unverifiable. Compare Test A with Test B and compare accuracy, completeness, reproducibility, efficiency, uncertainty, and evidentiary traceability across all four modalities. Explain disagreements, failed tests, hallucinations, and the effects of preprocessing. Support the comparison with at least one cross-modality table or figure and cite the corresponding appendix records. Report denominators and calculation methods for quantitative metrics; do not present an unsupported overall accuracy score.

4.9 Reflection and Discussion

Answer every required reflection question using evidence from the recorded tests. Discuss transparency limitations, privacy, security, unexpected agent actions, persuasive but incorrect output, tasks that must remain under examiner control, and safeguards necessary for real investigations. Explain how the hands-on work changed the team’s assessment of AI-agent usefulness in digital forensics. Organize the discussion around the nine required reflection questions, but synthesize related answers into a coherent argument rather than presenting unsupported opinions. Cite specific test IDs, screenshots, metrics, or appendix records as evidence.

4.10 Conclusion and Future Work

Summarize what the agent could and could not do reliably, threats to validity, limitations, and future work. Do not introduce new results. End with a clear judgment about appropriate forensic uses, tasks requiring direct examiner control, and the most important next evaluation step.

4.11 References

Use one consistent citation and reference style for papers, datasets, software, and documentation. Every in-text citation must appear in the reference list and every listed source must be cited. Verify authors, titles, venues, years, URLs, and DOIs; software, models, documentation, and datasets also require citations.

4.12 Appendices

Organize the supporting materials using the following lettered appendix headings. Refer to each appendix and test record from the main report. Appendices are excluded from the page limit but should remain organized, legible, and cross-referenced rather than used to avoid necessary analysis in the main report. The appendices should normally total 18–28 pages. This is a planning guideline, not a strict maximum. Each of the eight test records will generally require 2–3 pages when prompts, material output, validation, metrics, and legible screenshots are included. Longer appendices are acceptable when needed for readable evidence, but redundant screenshots and unfiltered logs belong in GitHub rather than the report.

4.12.1 Appendix A: Text Evidence Test Records

Include **Test A: Instructor Benchmark** and **Test B: Student-Selected Case**.

4.12.2 Appendix B: Image Evidence Test Records

Include **Test A: Instructor Benchmark** and **Test B: Student-Selected Case**.

4.12.3 Appendix C: Audio Evidence Test Records

Include **Test A: Instructor Benchmark** and **Test B: Student-Selected Case**.

4.12.4 Appendix D: Video Evidence Test Records

Include **Test A: Instructor Benchmark** and **Test B: Student-Selected Case**.

Each of the eight complete test records in Appendices A–D must follow the eight-field format demonstrated in Table 1. Include the complete sanitized prompt and material output, configuration and preprocessing, evidence and derived-file hashes, ground truth, validation commands and results, operational observations, required screenshots, and test conclusion. The *Multimodal Investigation Results* section must summarize and cite these records; the *Validation and Cross-Modality Comparison* section must analyze their validation results and claim classifications.

4.12.5 Appendix E: Team Member Contributions

Every member of the three- or four-student team must provide a self-report of individual contributions. Copy the following entry for each team member:

- Name and role.
- Technical, investigative, and writing contributions.
- Evidence modalities, tests, analysis, and report sections owned or supported.
- Challenges encountered and how they were addressed.
- Collaboration with other team members.

4.12.6 Appendix F: GitHub Reproducibility Repository

Provide the repository URL, final commit identifier, access status, and README location. Identify any evidence excluded because of size, privacy, licensing, safety, or redistribution restrictions and provide its filename, cryptographic hash, lawful source or instructor-access instructions, and authorized reconstruction or acquisition procedure. The GitHub repository must contain the complete sanitized reproducibility materials required by the Submission section.

5 Report Format and Collaboration Options

Teams may prepare the report using Microsoft Word, Google Docs, LaTeX, or another suitable document editor. No particular template or authoring system is required, and the choice of software does not affect the grade. Use a professional single-column layout with an 11- or 12-point readable font, approximately 1-inch margins, page numbers,

numbered headings, and consistently numbered and captioned figures and tables. Retain the required section and appendix structure and use one consistent citation and reference style throughout.

LaTeX users may work in Overleaf, OpenAI Prism, or a local editor, but they are not required to use the Coventry template or any other prescribed template. Word and Google Docs users may use their available shared-editing platforms. Verify that every team member can access the selected platform before beginning the report, and ensure that screenshots, tables, graphs, and other evidence remain legible in the exported PDF.

Every in-text citation must appear in the reference list and every listed source must be cited. Verify authors, titles, venues, years, URLs, and DOIs; software, models, documentation, and datasets also require citations. Whichever format is selected, the team must retain the editable report source, references, figures, tables, and final PDF. Platform AI features may assist writing or formatting, but students remain responsible for factual accuracy, original analysis, valid citations, and compliance with evidence-handling rules.

6 Project Requirements Summary

- **Group Size:** Three or four students per team; every member must complete the Team Member Contributions appendix.
- **Points:** 100 total; no extra-credit points are offered.
- **Papers:** Review and cite at least 10 relevant research papers.
- **Agent:** Install, operate, and evaluate either Hermes Agent or OpenClaw in isolation.
- **Coverage:** Conduct detailed testing of text, image, audio, and video evidence; no modality may be omitted.
- **Testing:** Complete benchmark Test A and approved student-selected Test B for every modality. Across the pair, include one normal and one adversarial, ambiguous, or failure-oriented task, and answer the common questions for every test.
- **Records:** Submit the required test records, sanitized prompts and outputs, analysis results, provenance, validation results, and reflection. Every test must include at least one legible screenshot, with additional validation screenshots when applicable.
- **GitHub:** Upload the complete sanitized reproducibility materials to one team GitHub repository, submit its URL and final commit identifier, and grant the instructor access if it is private.

- **Validation:** Compare results with ground truth or a conventional examination.
- **Safety:** Use copied, non-sensitive evidence with least privilege.
- **Report:** A professional technical report normally 25–30 pages, including references but excluding the title page, table of contents, and appendices; no particular authoring system or template is required.
- **Appendices:** Normally 18–28 pages for the eight complete test records, team contributions, and GitHub documentation; longer appendices are allowed when needed for readable evidence.
- **Presentation:** 10–15 minutes on the last day of class.

Table 3: *Rubric for Final Hands-On Investigation Project (100 Points)*

Category	Points	Performance Criteria
Literature Review	15	Excellent (full points): Synthesizes at least 10 relevant papers and applies their findings, risks, and gaps to the hands-on investigation. Good (75%): Uses at least 10 papers with generally clear connections but limited synthesis or application. Needs Improvement (50% or less): Uses fewer than 10 papers, weak sources, disconnected summaries, or little connection to the investigation.
Investigation Environment and Procedure	10	Excellent (full points): Concisely documents the isolated environment, versions, permissions, evidence, prompts, tools, preprocessing, provenance, ground truth, and validation procedure for all eight tests. Good (75%): Covers all eight tests with minor reproducibility or documentation omissions. Needs Improvement (50% or less): Presents an unsafe or non-reproducible procedure, unverified ground truth, or substantial omissions.

Continued on the next page

Table 3—continued from the previous page

Category	Points	Performance Criteria
Agent Installation and Configuration	15	Excellent (full points): Installs and operates Hermes Agent or OpenClaw in an isolated least-privilege environment; documents configuration, model and provider, enabled tools, troubleshooting, and observable actions. Good (75%): Provides a working isolated installation with minor configuration, logging, or troubleshooting gaps. Needs Improvement (50% or less): Provides a nonworking or unsafe installation that does not satisfy the required hands-on project.
Multimodal Investigation Execution	25	Excellent (full points): Completes benchmark Test A and approved student-selected Test B for text, image, audio, and video, including exact records, screenshots, and auditable agent actions. Good (75%): Completes all eight tests and all four modalities with only minor execution, record, or screenshot gaps. Needs Improvement (50% or less): Omits any modality or required case, or provides substantially incomplete or unauditable execution.
Validation, Results, and Reflection	20	Excellent (full points): Independently validates all eight tests; distinguishes supported, incorrect, unsupported, missing, ambiguous, and unverifiable claims; compares Test A, Test B, and all modalities; answers every reflection question with evidence. Good (75%): Validates and discusses all eight tests with minor gaps in metrics, comparison, limitations, or reflection. Needs Improvement (50% or less): Omits validation or test analysis, presents unsupported conclusions or an incomplete reflection, or weakly addresses risks and examiner oversight.

Continued on the next page

Table 3—continued from the previous page

Category	Points	Performance Criteria
Report Quality	10	Excellent (full points): Presents a clear professional technical report, normally 25–30 pages, with integrated results, citations, and analysis; includes complete, readable appendices, normally 18–28 pages, and an accessible GitHub repository containing eight test records and all required sanitized reproducibility materials. Good (75%): Provides a complete readable report with minor organization, formatting, citation, screenshot, or appendix issues. Needs Improvement (50% or less): Has major format or length problems, or is missing records, screenshots, analysis, citations, or reproducibility materials.
Presentation and Demonstration	5	Excellent (full points): Gives a clear 10–15 minute presentation covering all four modalities and a representative live or recorded agent demonstration. Good (75%): Covers all modalities and includes a demonstration, with minor clarity, organization, or timing issues. Needs Improvement (50% or less): Omits modality findings or the demonstration, is unclear, or substantially exceeds the time limit.
Total	100	

7 Evaluation Rubric

The project is graded according to the rubric in Table 3. Full criteria are provided for transparency. Hands-on installation, multimodal execution, validation, recorded evidence, reflection, and demonstration account for most of the grade; procedural documentation is graded for the forensic reproducibility it provides. Documented modality limitations do not reduce credit by themselves. A team can earn full credit when an agent lacks native support for a modality if the team uses a defensible, fully documented preprocessing pipeline, preserves provenance and derived-file hashes, completes the required tests, and evaluates the limitation honestly.

8 Resources

8.1 Required Agent Resources

- Hermes Agent: <https://github.com/NousResearch/hermes-agent>
- Hermes documentation: <https://hermes-agent.nousresearch.com/docs/>
- OpenClaw: <https://github.com/openclaw/openclaw>
- OpenClaw documentation: <https://docs.openclaw.ai/>

Use current official documentation, record the installed version or commit and model and provider dependencies, and review installation scripts before running them.

8.2 Required Benchmark and Student-Selected Evidence

For each modality, Test A uses the instructor-provided CYFI 725 AI-Agent Forensics Benchmark distributed in the same GitHub repository as the final-project assignment. Verify its supplied SHA-256 manifest before testing. The benchmark contains fictional synthetic text, image, audio, and video cases and common instructions; students do not receive the private instructor answer key. Test B uses evidence proposed by the team and approved by the instructor before agent access.

8.2.1 How to Access and Verify 2026F_benchmark

1. Open the instructor-provided GitHub repository for the CYFI 725 final project. Clone or download the repository using GitHub’s repository controls.
2. Navigate to `2026F_benchmark/student_package`. Do not use the worked example or ask an LLM to replace the benchmark evidence.
3. Preserve the downloaded `student_package` directory unchanged and create a separate working copy inside the team’s isolated investigation environment. The directory must contain `INSTRUCTIONS.md`, `CASE_MANIFEST.md`, `SHA256SUMS.txt`, and the `text`, `image`, `audio`, and `video` case directories.
4. Read the included instructions and case manifest before testing. From the student-package directory, verify the files on Linux with:

```
sha256sum -c SHA256SUMS.txt
```

Use an equivalent SHA-256 verification procedure on Windows or macOS. Record the verification result in the report or GitHub materials.

5. Preserve the verified originals as read-only evidence and analyze copies. Hash and document every derived file created by OCR, transcription, frame extraction, conversion, or other preprocessing.
6. Use all four benchmark cases as Test A:
 - Text: `text/case_text_01.txt`
 - Image: `image/case_image_01.png`
 - Audio: `audio/case_audio_01.wav`
 - Video: `video/case_video_01.mp4`

The private instructor answer key is not part of the student package.

7. If the GitHub directory is unavailable, the student package does not contain the listed files, or any hash fails, stop and contact the instructor. Do not use a fork, unofficial mirror, regenerated artifact, or other replacement, and do not continue with a failed integrity check.

The Test B proposal must identify the lawful source or creation method, investigative question, known ground truth, file format and size, SHA-256 hash, privacy and copyright considerations, planned conventional validation, and any generative-AI assistance. Public URLs alone are insufficient because content may change or disappear; preserve an approved local copy when licensing permits. Synthetic evidence is acceptable, but the team must independently verify its contents and may not treat another model's output as ground truth.

8.3 Forensic Evidence and Datasets

- Course lecture USB disk image: download `usb_dd.7z`. Extract and analyze only a verified working copy; calculate its hash and compare it with the hash distributed with the image before testing.
- NIST Computer Forensic Reference Data Sets: <https://www.nist.gov/itl/ssd/software-quality-group/computer-forensic-reference-data-sets>
- Digital Corpora: <https://digitalcorpora.org/>

8.4 Example Research Papers

- “A Framework for Integrated Digital Forensic Investigation Employing AutoGen AI Agents” [1].
- “ChatGPT for Digital Forensic Investigation: The Good, The Bad, and The Unknown” [2].

- “LLM-driven Provenance Forensics for Threat Investigation and Detection” [3].
- “Foundations for Agentic AI Investigations from the Forensic Analysis of Open-Claw” [4].
- “LLM-Powered Automated Cloud Forensics: From Log Analysis to Investigation” [5].

8.5 Example Investigation Questions

- Can the agent reconstruct an accurate, source-traceable timeline from logs?
- Can it interpret artifacts without altering evidence or inventing facts?
- Can it extract IOCs as accurately as a conventional examination?
- How do findings differ between Test A and Test B within each modality?
- Does it identify uncertainty and missing evidence, or hallucinate a conclusion?

9 Notes for Students

- Start early; installation, model access, isolation, and testing may require troubleshooting.
- Keep a test log with versions, settings, prompts, actions, outputs, validation, and unexpected behavior.
- Use the required test-record fields for every text, image, audio, and video test. Include key result tables, representative screenshots, and analysis in the main report, which should normally be 25–30 pages. A modest extension is acceptable for readable evidence and substantive analysis. The title page, table of contents, and appendices do not count toward this guideline. Appendices should normally total 18–28 pages; place complete records and supporting screenshots there, while storing unfiltered logs and additional materials in GitHub.
- Sanitize submissions; never include keys, tokens, passwords, personal information, or sensitive evidence.
- Maintain one team GitHub repository for sanitized reproducibility materials. Use a private repository if public release would violate evidence, privacy, licensing, or course restrictions; grant the instructor access before submission.
- Use IEEE Xplore, ACM Digital Library, or arXiv and verify bibliographic details.

- Use Word, Google Docs, LaTeX, or another suitable editor. No particular template is required; retain the complete editable report source in every format.
- Practice the demonstration and explain ground truth, errors, and limitations.

10 Submission

Submit the report, normally 25–30 pages, as a PDF, together with its editable source file or an instructor-accessible link to the editable source, and optional presentation slides to Canvas. The paper must contain key result tables, screenshots, and analysis. Submit the team GitHub repository URL and the final commit identifier through Canvas and include both in the report appendix. The repository must contain the complete sanitized test records, prompts and outputs, supporting screenshots, configuration and environment documentation, scripts and commands, evidence and derived-file hashes, preprocessing records, validation materials, and a README explaining how to reproduce the observable workflow. Do not commit API keys, tokens, credentials, personal data, confidential material, malware, or evidence that the team is not permitted to redistribute. For excluded or oversized evidence, include the filename, cryptographic hash, lawful source or instructor-access instructions, and the documented procedure needed to reconstruct or obtain the authorized working copy; never bypass GitHub or licensing restrictions.

11 Conclusion

This project develops practical skills in deploying, using, documenting, and critically evaluating AI agents in digital forensic investigations. By deploying Hermes Agent or OpenClaw in isolation, testing it against known evidence, and auditing both correct and unreliable behavior, students will learn where agentic automation may assist an examiner and where professional judgment remains essential. Reproducibility, traceability, evidence protection, and independent validation remain central.

References

- [1] M. Al Fahdi, N. L. Clarke, F. Li, and H. Al-Mohannadi, “A Framework for Integrated Digital Forensic Investigation Employing AutoGen AI Agents,” in *Proc. 12th Int. Symp. Digital Forensics and Security*, 2024, doi: 10.1109/ISDFS60797.2024.10527235.
- [2] M. Scanlon, F. Breiteringer, C. Hargreaves, J.-N. Hilgert, and J. Sheppard, “ChatGPT for Digital Forensic Investigation: The Good, The Bad, and The Unknown,” arXiv:2307.10195, 2023. [Online]. Available: <https://arxiv.org/abs/2307.10195>.

- [3] K. Mukherjee and M. Kantarcioglu, “LLM-driven Provenance Forensics for Threat Investigation and Detection,” arXiv:2508.21323, 2025. [Online]. Available: <https://arxiv.org/abs/2508.21323>.
- [4] J. Gruber and J.-N. Hilgert, “Foundations for Agentic AI Investigations from the Forensic Analysis of OpenClaw,” arXiv:2604.05589, 2026. [Online]. Available: <https://arxiv.org/abs/2604.05589>.
- [5] “LLM-Powered Automated Cloud Forensics: From Log Analysis to Investigation,” in *Proc. IEEE 18th Int. Conf. Cloud Computing*, 2025, doi: 10.1109/CLOUD67622.2025.00012.